Strategy Espresso: EU–India Free Trade Agreement & European Equities

- **A landmark deal two decades in the making.** The EU and India have reached political agreement on a Free Trade Agreement covering ~2bn people and ~25% of global GDP — a major realignment in a world of rising protectionism and renewed tariff uncertainty from the US.
- **Macro impact modest near term.** Our economists estimate that even a fully implemented FTA will not offset the export drag from higher US tariffs, underscoring that diversification helps but cannot fully substitute for US demand.
- **But meaningful sector-level gains.** India will eliminate or sharply reduce tariffs on 96.6% of EU goods exports — particularly autos, machinery, chemicals and aerospace — unlocking real opportunities in the most tariff-constrained sectors.
- **European companies exposed to India (GSSTINDI) are on the rise.** Our GSSTINDI basket is the best-performing regional exposure YTD (+6.2%), offering access to India's growth at a reasonable valuation. These companies deliver market-level top-line growth at a 7% discount, and on a PEG basis trade on a deep discount vs history, with consensus expecting a 10–15% earnings CAGR.
- **India, a structural grower.** Our Asia economists expect India's GDP to grow 6.7% in 2026 and 6.8% in 2027 (above consensus) with inflation returning toward target — and our FX strategists see INR past the worst of its underperformance. This provides a positive backdrop for international companies exposed into India. We reiterate our preference for diversification – increasing exposure to Emerging Markets.

Guillaume Jaisson
+44(20)7552-3000 |
guillaume.jaisson@gs.com
Goldman Sachs International

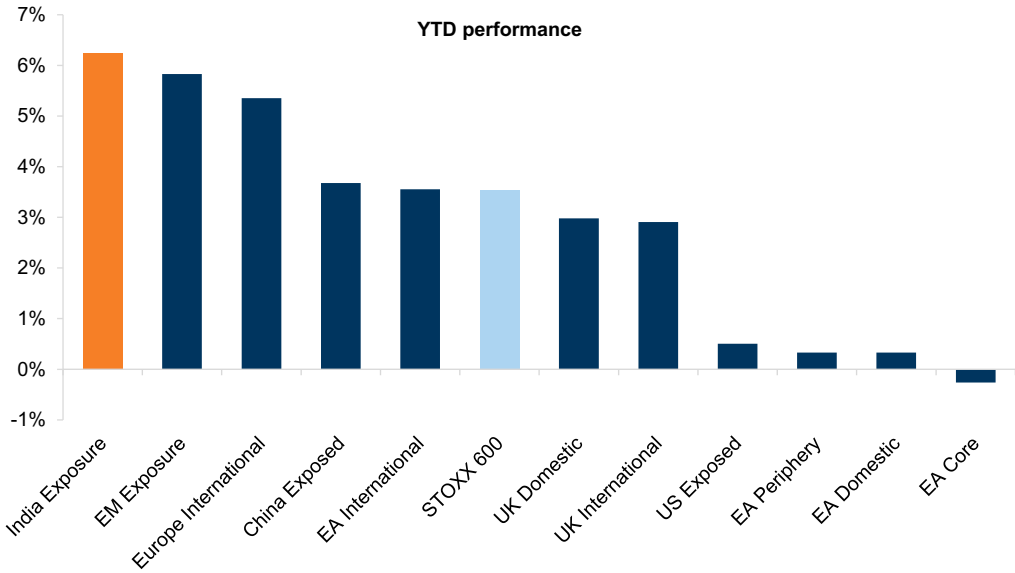
Peter Oppenheimer
+44(20)7552-5782 |
peter.oppenheimer@gs.com
Goldman Sachs International

Sharon Bell
+44(20)7552-1341 |
sharon.bell@gs.com
Goldman Sachs International

Giovanni Ferrannini
+44(20)7051-2589 |
giovanni.ferrannini@gs.com
Goldman Sachs International

Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html.

Exhibit 1: Our basket of European companies exposed to India has delivered the best performance YTD, among a selection of geographical exposure baskets



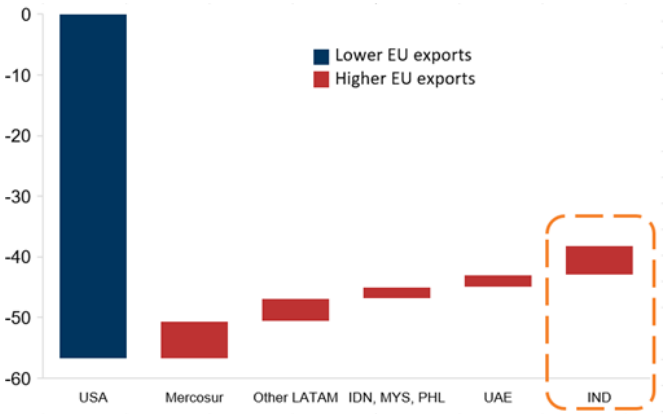
Source: Bloomberg, Goldman Sachs Global Investment Research

'The mother of all deals'

The "mother of all deals". India and Europe are recalibrating trade and strategic relationships amid rising global protectionism. The EU–India Free Trade Agreement (FTA) concludes one of the longest trade negotiations on record. Talks began in 2007 and just reached political agreement, after almost two decades of intermittent progress. Formal signing is expected later in 2026, following legal vetting and ratification by EU member states, the European Parliament and the Council. Leaders have labelled it the "mother of all deals", reflecting both its economic ambition and its role as a strategic response to a more coercive global trade environment. India (~1.5bn people) and the EU (~450m) together account for ~2bn people, ~25% of global GDP, and around one quarter of the world's population and trade. Similarly, the UK is using bilateral agreements to anchor its post Brexit trade strategy. This realignment has been accelerated by US policy uncertainty, including 50% tariffs imposed on India in 2025 and renewed tariff risks for Europe, pushing all sides to diversify trade and supply chain exposure.

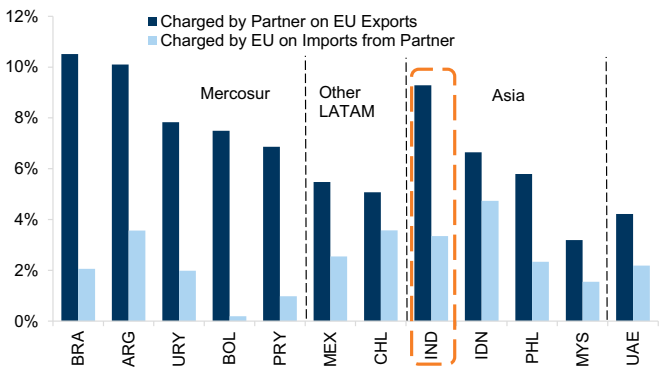
The macro impact remains modest near term. Our European Economists highlight that even a fully fledged set of new EU trade agreements would raise EU gross exports by <€20bn (≈0.1% of GDP), covering only around one-third of the expected export shortfall caused by higher US tariffs (Exhibit 2). This underscores that Europe's external demand uplift from diversification is real but limited in the short term horizon. Our economists also emphasise that while this FTA accelerates an already deep relationship, Europe's overall export performance remains constrained by rising global protectionism, heightened Chinese competition, and persistently high US tariffs.

Exhibit 2: Stylised impact of tariff developments on EU exports



Source: Comtrade, IMF, WTO, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 3: Bilateral Effective Tariff Rates*



* Based on 2023 MFN rates, weighted by 2024 trade flows.

Source: Comtrade, IMF, WTO, Haver Analytics, Goldman Sachs Global Investment Research

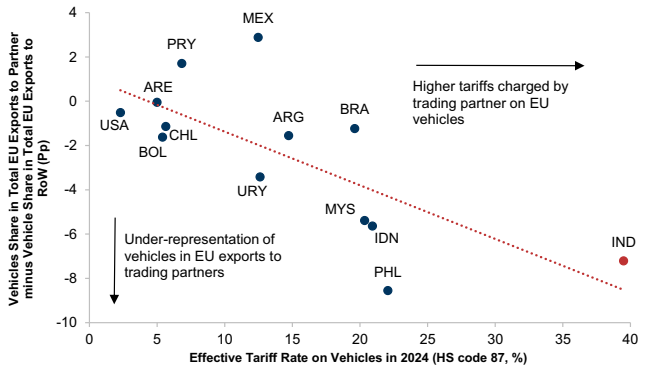
Trade terms. The agreement will eliminate or reduce tariffs on 96.6% of EU goods exports to India (Exhibit 3 above), generating estimated savings of ~€4bn per year for EU exporters. Indian tariffs will fall to zero over time for large parts of machinery, medical equipment (previously up to 44%), chemicals (22%), pharmaceuticals (11%), plastics (16.5%) and aerospace (11%). Auto tariffs will drop from 110% to 10% under a quota of 250,000 vehicles; auto parts will be fully duty free 5–10 years after entry into force. Wine tariffs will fall from 150% to 20–30%, spirits from 150% to 40%, beer from 110% to 50%, and olive oil from 45% to zero after five years. To secure political support in

Europe, the EU retains protection for sensitive agricultural sectors including beef, poultry, rice and sugar, while dairy is excluded on the Indian side. Strict rules of origin limit preferential access to goods genuinely produced within the two blocs, preventing trade diversion via third countries.

Meanwhile, the EU's Carbon Border Adjustment Mechanism (CBAM) charges a carbon price on the embedded direct emissions of select imports of Iron & Steel, Aluminium, Cement, Fertilisers, Hydrogen and Electricity. Our analysts, believe CBAM will help provide import protection by applying similar standards of pricing emissions to imports as it does to domestic suppliers. They estimate charges by country suggesting Indian Steel and Chinese Aluminium imports would be most impacted if CBAM costs are borne by producers.

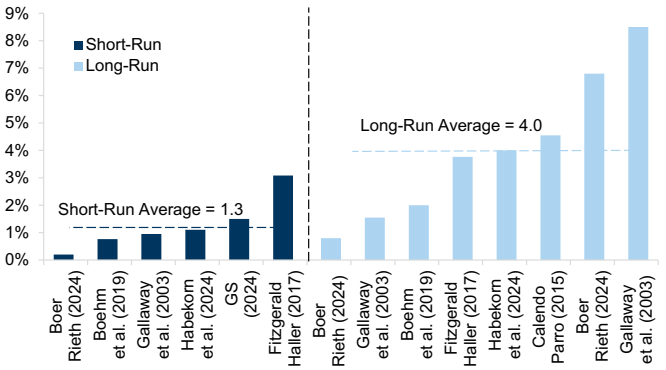
Industrials are the main beneficiaries. Our economists note that the largest short-term beneficiaries are likely to be EU autos, chemicals and electrical machinery — sectors that currently face the highest foreign tariff barriers and where EU exports remain under-represented in key Asian markets (Exhibit 4 below). The literature points to a trade elasticity to the tariff rate of about 1.3 and 4 in the short and long run, respectively (Exhibit 5 below). So a 10% decline in tariffs boost exports by 13% over the short-run and 40% over the long-run. We note that European Autos & Parts have outperformed their Indian peers as news of the FTA has been reported in the press recently (8% outperformance in the last 2 weeks).

Exhibit 4: Protectionism in Autos Sector



Source: Comtrade, IMF, WTO, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 5: Estimates of Trade Elasticities*

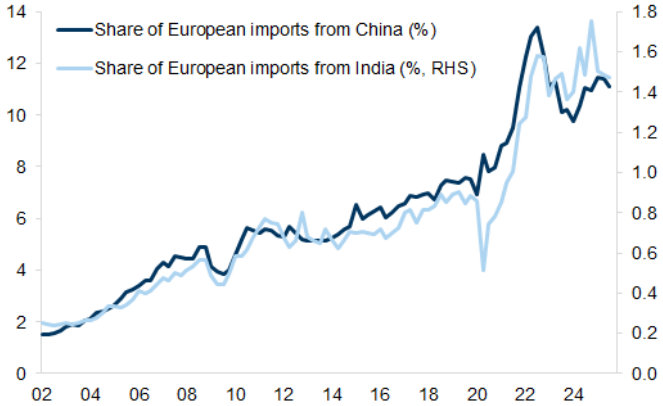


*based on 2023 MFN rates, weighted by 2024 trade flows

Source: Comtrade, IMF, WTO, Haver Analytics, Goldman Sachs Global Investment Research

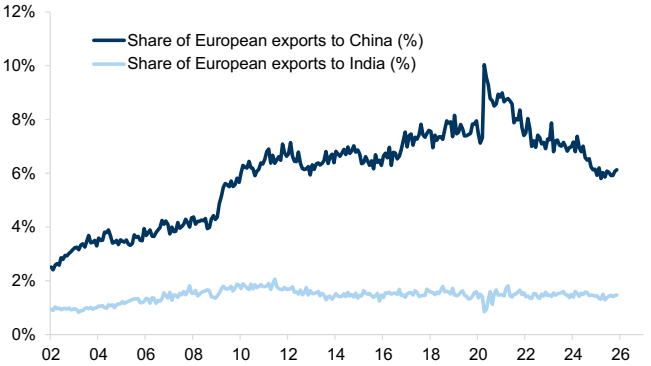
Scaling an existing relationship. The EU is already India's largest trading partner, with bilateral trade reaching $136.5bn in India's fiscal year through March 2025, accounting for >17% of India's total exports. EU goods exports to India are concentrated in machinery, transport equipment and chemicals, while services integration has deepened materially: Indian services exports to the EU rose from around €20bn in 2019 to around €40bn in 2024, and EU services exports to India reached around €30bn. The Commission expects EU goods exports to India to double by 2032, underscoring that the FTA is about acceleration rather than creation.

Exhibit 6: Even though the magnitude of Europe's imports from India are much lower currently, they follow a similar trend as that of China
Share of European Imports (%)



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 7: Taking China as a precedent, we see scope for European companies to increase their customer base in India
Share of European exports (%)



Source: Haver Analytics, Goldman Sachs Global Investment Research

Potential to scale. The trade pact is accompanied by a new EU–India security partnership, offering closer cooperation on maritime security, space, hybrid threats and cyber attacks, and opening the door to expanded defence industry collaboration. Climate cooperation remains anchored in the Clean Energy and Climate Partnership, while the agreement also introduces frameworks for labour mobility.

Commonwealth of Nations. In parallel, the UK concluded its own landmark agreement with India in 2025, its largest and most economically significant trade deal since Brexit. Average tariffs on UK exports to India fall from 15% to 3%.

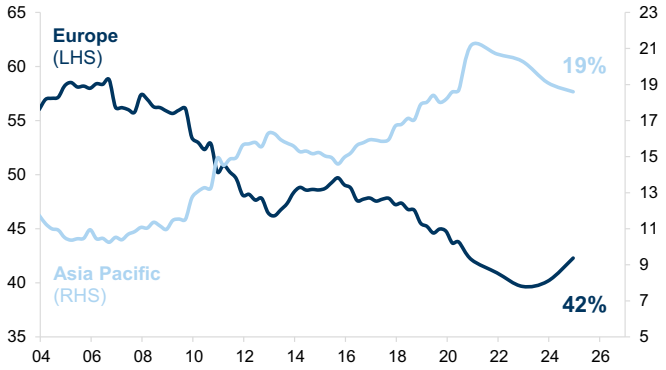
Diversification as strategy. Politically, these agreements signal a shared preference for trade agreements over tariffs at a time of heightened uncertainty. Strategically, they reflect India's push to shed its protectionist image and diversify export markets, Europe's effort to "de-risk" away from the US and China, and the UK's attempt to re-establish global trade relevance post Brexit. Together, they underline India's rising centrality in a fragmenting global trading system where diversification has become the dominant strategic objective.

India is the most populous country globally. Our Asia Economist expect India's real GDP growth at 6.7% in 2026 and 6.8% in 2027—above consensus—with bank credit growth re-accelerating towards ~13% y/y and both rural and urban consumption strengthening. The team highlights a decisive "policy put" implemented in 2025–26: a 125bp RBI easing cycle, extensive liquidity injections (~4.7% of NDTL), income-tax and GST cuts, and targeted regulatory relaxation. Our economists estimate that India's GDP can grow above 6% pa over the next decade and that it will overtake the US in 50 years. Moreover, India is focused on becoming a major manufacturing alternative to China. The government is investing heavily to develop globally competitive manufacturing hubs as the world restructures its supply chains.

Global companies are seeking to take advantage of this potential growth opportunity. European companies are largely international: the STOXX 600 derives only 42% of its revenues from Europe, with the majority from the rest of the world: 24% from North America, 19% from Asia-Pacific and 15% from other Emerging Markets (Exhibit 8). That said, only a small portion of its revenues come from India (<2%).

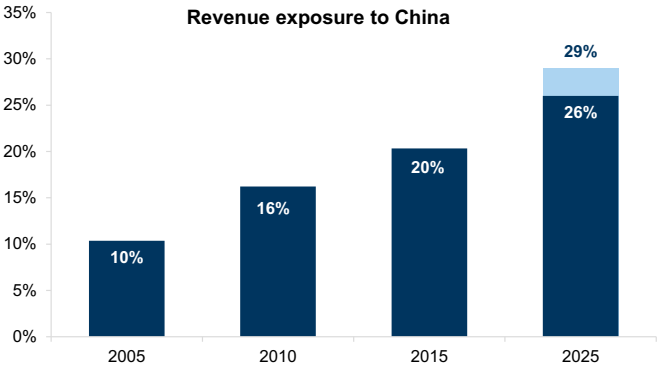
Our basket of European companies exposed to India (GSSTINDI) is the best performing basket year to date (Exhibit 10). GSSTINDI is up 6.2% outperforming the market (up 3.5%) and also European companies exposed to China (+3.7%) or Emerging Markets (+5.8%). We recognize that, direct exposure to India remains limited; but we think it is a first milestone to get exposure to India. Taking China exposure as a precedent, we see some scope for European companies to increase their exposure to India. Our basket of European companies exposed to China (GSSTCHNA) generated only 10% of its revenues from China in 2005, but 20% in 2015 and almost 30% today (Exhibit 9).

Exhibit 8: The impact of globalisation on STOXX Europe 600
STOXX Europe 600 – Geographic Revenue Exposure



Source: FactSet, STOXX, Goldman Sachs Global Investment Research

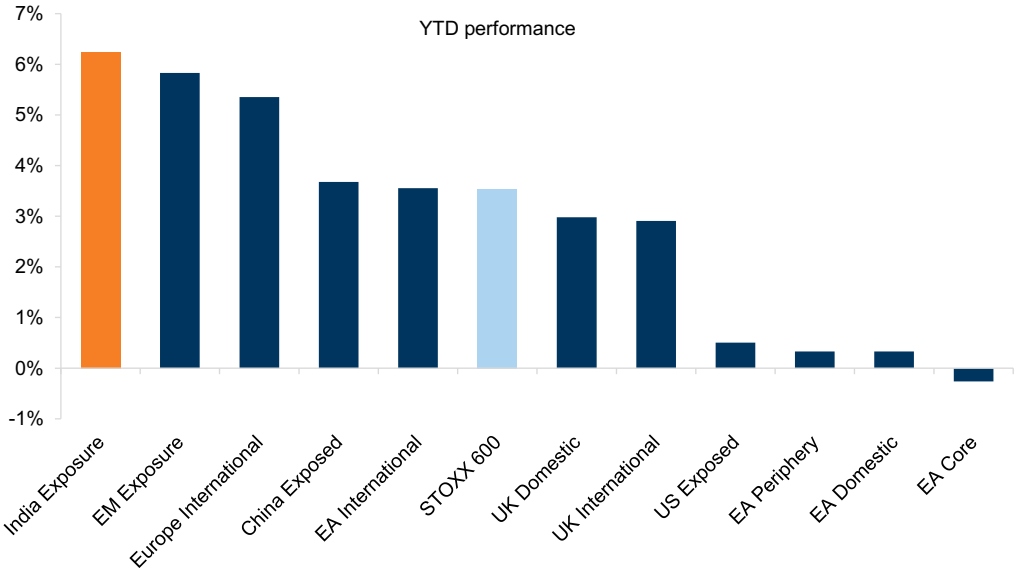
Exhibit 9: European companies exposed to China (GSSTCHNA) only generated 10% of their revenues from China in 2005, 20% in 2015 against 29% today
European companies exposed to Greater China (GSSTCHNA). Revenue exposure over time



Adjustment made in 2025 for European Autos joint ventures in Greater China and for European Luxury sales to Chinese nationals travelling abroad

Source: FactSet, Goldman Sachs Global Investment Research

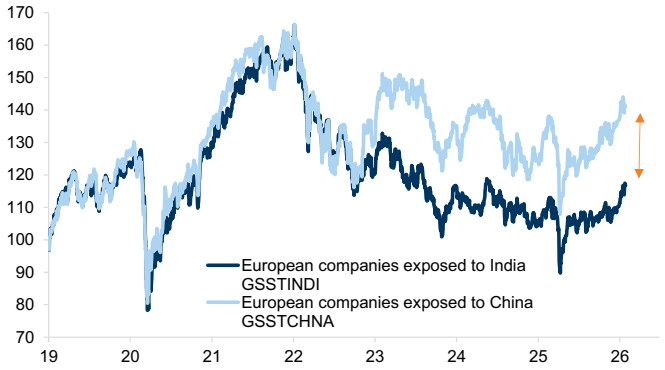
Exhibit 10: Our basket of European companies exposed to India has delivered the best performance YTD, among a selection of geographical exposure baskets



Source: Bloomberg, Goldman Sachs Global Investment Research

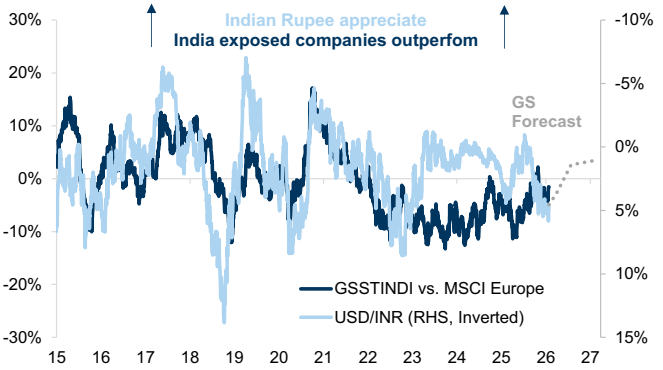
GSSTINDI constituents are positively correlated with GSSTCHNA (Exhibit 11). The two baskets have some overlap, like most baskets exposed to Emerging Markets. GSSTINDI constituents have tended to outperform the market when the Indian Rupee appreciates (Exhibit 12). Our Indian Economists expect India headline inflation at ~3.9% in 2026, close to target, implying limited scope for further rate cuts unless US-related tariff uncertainty persists. The INR is seen by our economists as past its worst underperformance, with USD/INR at 92/93/94 over the next 3/6/12 months in their baseline—providing a more stable macro backdrop for foreign corporates expanding into India.

Exhibit 11: European companies exposed to India (GSSTINDI) are positively correlated with European companies exposed to China (GSSTCHNA)
Price performance (indexed to 100 in Jan-2019)



Source: Datastream, Goldman Sachs Global Investment Research

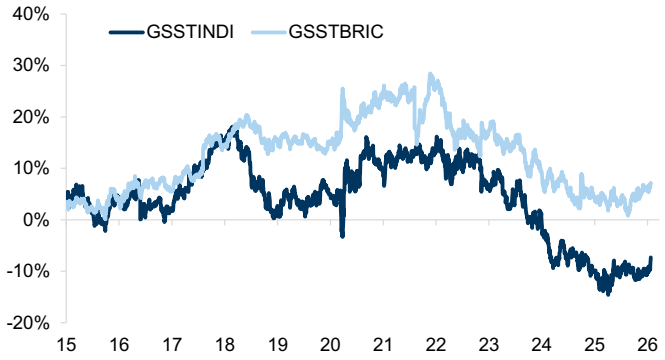
Exhibit 12: GSSTINDI has historically tended to outperform when the Indian Rupee appreciates
Relative price performance over 6m



Source: Datastream, Goldman Sachs Global Investment Research

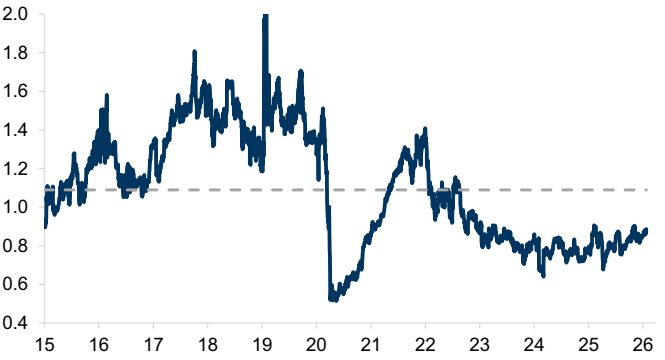
GSSTINDI offers access to this growth at a reasonable price, in our view. The basket trades on 15x P/E and in line with its historical average, while MSCI India trades on 21x P/E, its 80th percentile since 2010. Exhibit 15 shows the 'Value of Growth' for each sector in Europe, alongside some EM exposed slices of the European market. The stronger the growth (on the x-axis), the higher the valuation (on the y-axis). On this basis, the valuation of our basket of European companies exposed to India (GSSTINDI) does not look excessive given the growth outlook it offers. Basically, these stocks offer the same top-line growth as the market but are on a 7% discount (Exhibit 13). Similarly, on a PEG ratio basis, GSSTINDI trades on a discount vs history (Exhibit 14) - as consensus expects 10% to 15% earnings CAGR.

Exhibit 13: Relative PE Premium/Discount of GSSTINDI and GSSTBRIC vs. STOXX 600
GSSTINDI = European companies exposed to India; GSSTBRIC = European companies with EM exposure



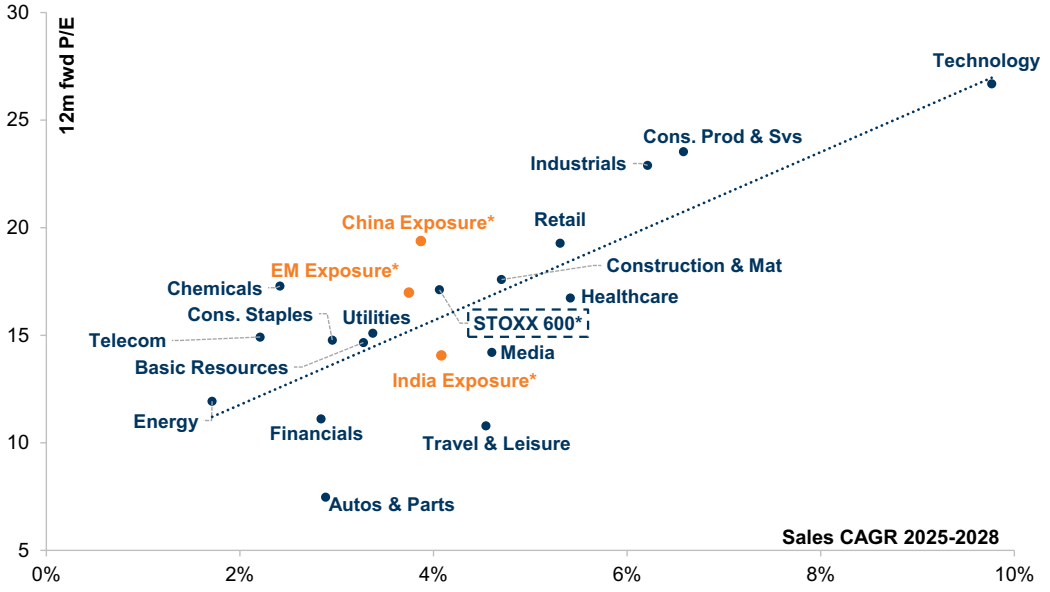
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 14: GSSTINDI offers growth at a reasonable price
PEG ratio (12m fwd P/E divided by second 12m EPS growth) of European companies with revenues exposure to India (GSSTINDI). Historical median in grey



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 15: GSSTINDI screens as inexpensive vs. the market, relative to its growth prospects
12m fwd P/E and 2028/2025 Sales CAGR (%)

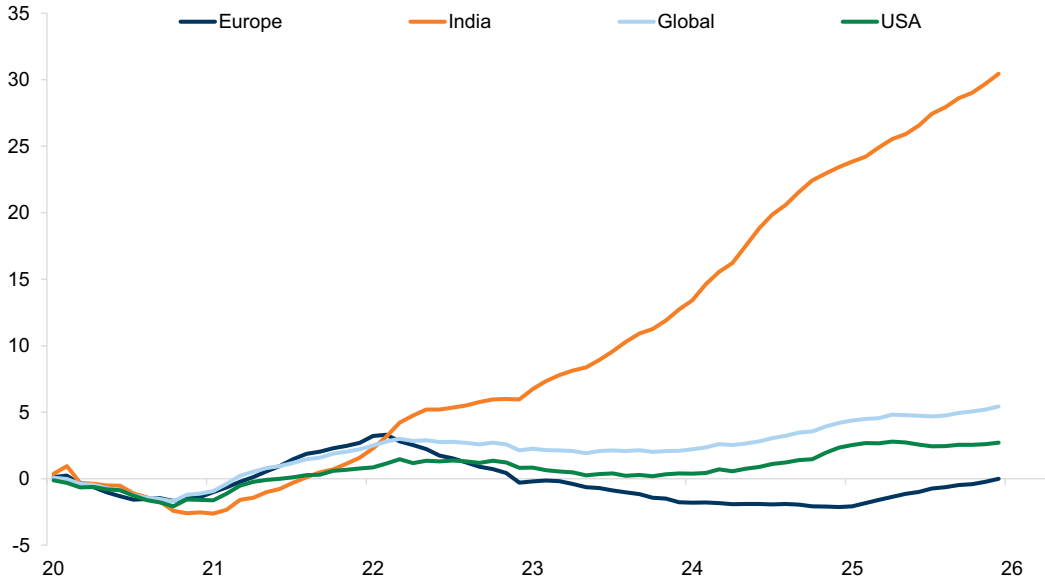


* baskets and indexes ex financials. China exposure: GSSTCHNA; EM exposure: GSSTBRIC; India Exposure: GSSTINDI – see our Portfolio Passport, Dec 2025

Source: Datastream, FactSet, Goldman Sachs Global Investment Research

Flows. Indian equities have attracted significant inflows recently. The assets under management we track with EPFR increased by 30% since 2020 (Exhibit 16), while they remained flat in other parts of the world and decreased in Europe.

Exhibit 16: Since 2020, Indian equities have attracted significant inflows while other parts of the world have seen outflows
Cumulative Flows as a % of AUM



Source: EPFR, Goldman Sachs Global Investment Research

Appendix

Key criteria: Companies with high sales exposure to India – *see: European companies exposed to India: The Nifty Economy, 6 July 2023.*

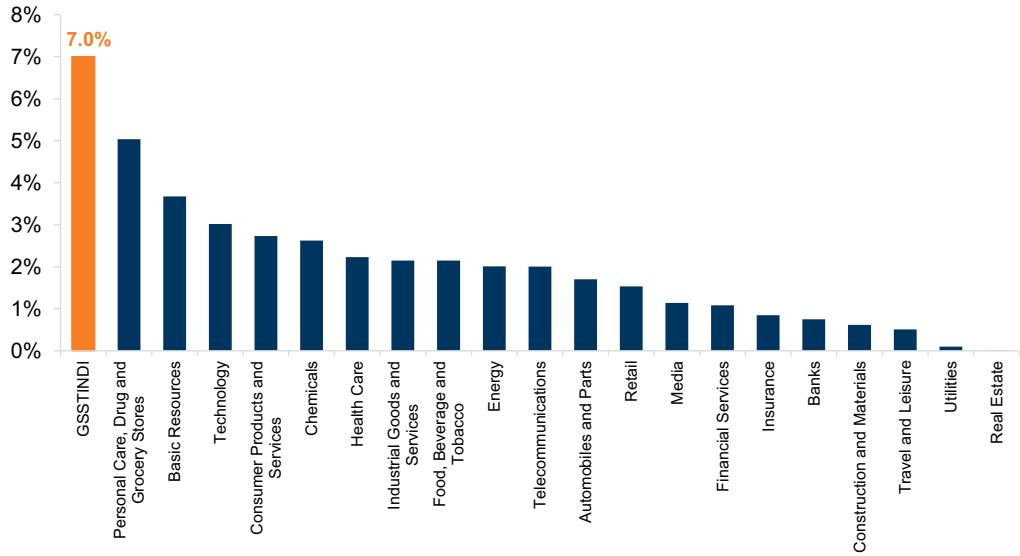
Sector exposure: Industrial Goods and Services 38%; Technology 14%.

Exhibit 17: Indian Exposure Basket Constituents
FY-2024 Sales Exposure

Indian Exposure (GSSTINDI)													
Company name	Country	Basket weights	India exposure	APAC exposure	Market Cap (EUR Bn)	NTM P/E	Company name	Country	Basket weights	India exposure	APAC exposure	Market Cap (EUR Bn)	NTM P/E
Automobiles and Parts		**6.9%**					Alfa Laval	Sweden	3.4%	7%	41%	20.0	22.8
CIE Automotive	Spain	3.4%	17%	24%	3.5	9.2	Wartsila	Finland	3.4%	5%	30%	19.5	28.7
Forvia	France	3.4%	5%	27%	2.9	7.4	Flughafen Zuerich	Switzerland	3.4%	6%	6%	8.1	22.3
Banks		**3.4%**					Teleperformance	France	3.4%	6%	12%	3.3	4.1
Standard Chartered	United Kingdom	3.4%	7%	65%	47.6	10.6	Huhtamaki	Finland	3.4%	6%	20%	3.3	11.8
Basic Resources		**6.9%**					INFICON Holding	Switzerland	3.4%	7%	49%	3.3	34.8
Glencore	United Kingdom	3.4%	7%	50%	67.6	17.4	Coats Group	United Kingdom	3.4%	12%	67%	1.9	11.2
SKF	Sweden	3.4%	5%	29%	10.8	13.8	**Personal Care, Drug and Grocery Stores**		**3.4%**				
Chemicals		**6.9%**					Unilever	United Kingdom	3.4%	11%	36%	122.2	17.2
Clariant	Switzerland	3.4%	6%	29%	2.7	9.0	**Technology**		**13.8%**				
RHI Magnesita	Netherlands	3.4%	13%	33%	1.5	6.7	Prosus	Netherlands	3.4%	11%	70%	261.4	12.3
Construction and Materials		**6.9%**					Infineon Technologies	Germany	3.4%	5%	61%	54.7	23.4
Signify	Netherlands	3.4%	5%	20%	2.7	7.7	Technoprobe	Italy	3.4%	10%	47%	17.1	59.9
Ariston Holding	Italy	3.4%	5%	12%	1.8	11.9	Nordic Semiconductor	Norway	3.4%	16%	77%	2.3	55.4
Food, Beverage and Tobacco		**6.9%**					**Telecommunications**		**6.9%**				
Diageo	United Kingdom	3.4%	7%	15%	42.5	13.7	Nokia	Finland	3.4%	14%	24%	33.2	17.7
Pernod Ricard	France	3.4%	13%	42%	19.0	12.4	LM Ericsson	Sweden	3.4%	6%	24%	30.2	13.8
Industrial Goods and Services		**37.9%**											
Airbus	France	3.4%	7%	26%	160.4	25.2							
Schneider Electric	France	3.4%	11%	27%	134.1	23.3							
Legrand	France	3.4%	5%	16%	33.7	22.4							
Dassault Aviation	France	3.4%	5%	13%	25.6	19.5							
Median			**7%**	**29%**	**19.0**	**13.8**							

Source: FactSet, Datastream, Goldman Sachs Global Investment Research

Exhibit 18: Sales exposure to India
European sectors and baskets



Source: FactSet, Goldman Sachs Global Investment Research

We would like to thank Madhura Balakumar and Jacinta Feng for their contributions to this report.

Europe Equity Strategy Team

Peter Oppenheimer
+44(20)7552-5782
peter.oppenheimer@gs.com
Goldman Sachs International

Sharon Bell
+44(20)7552-1341
sharon.bell@gs.com
Goldman Sachs International

Guillaume Jaisson
+44(20)7552-3000
guillaume.jaisson@gs.com
Goldman Sachs International

Giovanni Ferrannini
+44(20)7051-2589
giovanni.ferrannini@gs.com
Goldman Sachs International

Disclosure Appendix

Reg AC

We, Guillaume Jaisson, Peter Oppenheimer, Sharon Bell and Giovanni Ferrannini, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Marquee disclosure

Marquee is a product of the Goldman Sachs Global Markets Division. Any Marquee content linked in this report is not necessarily representative of GS Research views.

MSCI Disclosure

All MSCI data used in this report is the exclusive property of MSCI, Inc. (MSCI). Without prior written permission of MSCI, this information and any other MSCI intellectual property may not be reproduced or redisseminated in any form and may not be used to create any financial instruments or products or any indices. This information is provided on an "as is" basis, and the user of this information assumes the entire risk of any use made of this information. Neither MSCI, any of its affiliates nor any third party involved in, or related to, computing or compiling the data makes any express or implied warranties or representations with respect to this information (or the results to be obtained by the use thereof), and MSCI, its affiliates and any such third party hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any of this information. Without limiting any of the foregoing, in no event shall MSCI, any of its affiliates or any third party involved in, or related to, computing or compiling the data have any liability for any direct, indirect, special, punitive, consequential or any other damages (including lost profits) even if notified of the possibility of such damages. MSCI and the MSCI indexes are service marks of MSCI and its affiliates. The Global Industry Classification Standard (GICS) were developed by and is the exclusive property of MSCI and Standard & Poor's. GICS is a service mark of MSCI and S&P and has been licensed for use by The Goldman Sachs Group.

Basket disclosure

The ability to trade the basket(s) in this report will depend upon market conditions, including liquidity and borrow constraints at the time of trade.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf, and a copy of the annual audit

compliance report can be found at this link: https://publishing.gs.com/content/site/india-annual-compliance-report.html. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, The Investment Trusts Association, Japan, and Japan Investment Advisers Association. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the

investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.